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Density

We start off by considering some topological space X, endowed with a topology we will call τ_X. Then for some $A \subset X$, we have the following notion of density:

Definition 0.1. *A subset of a topological space $A \subseteq X$ is dense in X if (we propose two equivalent conditions):*

1. $\forall x \in X, \forall U_x \in \tau_X : U \cap A \neq \varnothing$
2. $\bar{A} = X$

Intuitively, no matter how close we look to a point in X, we can find a point in A, so A "covers" (perhaps an interesting adjective and intuively useful is "spans") all of X. We call this variety of dense as **everywhere dense**.

It however is possible that the closure of A is not ALL of X, but rather a subset of X. We will call this dense in a subset:

Definition 0.2. *Consider $A \subseteq Y \subseteq X$, then A is dense in Y if $Y \subseteq \bar{A}$.*

Note that we did not give the open set definition; for the purposes of further content dealing with closures is easier. Examining the condition $Y \subseteq \bar{A}$, we see that implies by definition of closure that $\bar{Y} \subseteq \bar{A}$. Furthermore, since $A \subseteq Y$, we have that $\bar{A} = \bar{Y}$. This is an equivalent condition, which will be indispensable moving forward, and we will default to this. Additionally, using the open set condition for dense in a subset requires an additional notion:

Definition 0.3. *The subspace topology is an induced topology on a subset $Y \subseteq X$, where:*

$$\tau_Y \coloneqq \{O \cap Y : O \in \tau_X\}$$

An open set relative to Y (we call this relatively open as well), is open if and only if it can be represented as the intersection of an open set $O \subseteq X$ and Y. We talked about A being dense in a subset meaning the closure of A is the closure of the subset. The closure we referred to was the closure of the sets with respect to the entire space X.

Consider the set $\mathbb{Q}$ in $\mathbb{Q}$ versus $\mathbb{Q}$ in $\mathbb{R}$. Then the closure of $\mathbb{Q}$ in the first case is $\mathbb{Q}$ itself; whereas, the closure in $\mathbb{R}$, is $\mathbb{R}$. Then:

Lemma 0.1. *The closure of A with respect to Y, which we will denote as $\overline{A}^Y$ is the same as $\bar{A} \cap Y$.*

Proof. We know that the definition of closure with respect to X is:

$$\bar{A} \coloneqq \bigcap \{C : C \text{ closed} \wedge A \subseteq C\}$$

Then for all such C which are closed subsets of Y, there exists some $B \subseteq X$ such that B is closed and $C = B \cap Y$. Then:

$$\bar{A}^Y = \bigcap C = \bigcap (B \cap Y) = \left(\bigcap B\right) \cap Y = \bar{A} \cap Y$$

□

Lemma 0.2. *For A dense in a subset Y, another equivalent form of the statement is $\bar{A}^Y = Y$.*

Proof. We first prove that $\bar{A} = \bar{Y}$ yields the equivalent statement. Simply take the intersection with Y for both sets such that:

$$\bar{A} \cap Y = \bar{A}^Y = \bar{Y} \cap Y = Y$$

Now if we have that $\bar{A}^Y = Y$, we can write it as $\bar{A} \cap Y = Y$. Thus:

$$Y = \bar{A} \cap Y \subseteq \bar{A} \implies \bar{Y} \subseteq \bar{A}$$

By subset properties once more, we get $\bar{A} = \bar{Y}$. □

Notice how in Definition 0.2, we required $A \subseteq Y$. We can actually weaken this condition to simply a nonempty intersection:

Definition 0.4. *A is* ***somewhere dense*** *if there exists some open $Y \subseteq X$ such that $\overline{A \cap Y} = \bar{Y}$*

Consider some open set Y such that $O \coloneqq A \cap Y$ and $A \cap Y \neq \varnothing$. Then $O \subseteq Y$, so if O is dense in Y:

$$\bar{O}^Y = Y \iff \bar{O} = \bar{Y} \iff \overline{A \cap Y} = \bar{Y}$$

This property of somewhere dense means part of A can span some subset, not necessarily the whole set. We give a few facts without apology:

1. A subset A is everywhere dense (we will refer to this as simply dense) if and only if somewhere dense for all open subsets Y of X
2. A subset A is somwhere dense if and only if $\text{int}(\bar{A}) \neq \varnothing$

We introduce the notion, jumping off of somewhere dense, of nowhere dense.

Definition 0.5. *A subset A is nowhere dense if and only if it is NOT somewhere dense.*

Notice by the (1) in the previous list, not dense **DOES NOT** imply nowhere dense. If a set is not dense, then there exists some Y in which it is not somewhere dense (but the lack of somewhere density is not necessarily true for all such open subsets).

However, a nowhere dense subset is not dense. By (2) we see that a subset A is nowhere dense if and only if $\text{int}(\bar{A}) = \varnothing$. This will be the main criterion for proving nowhere density as we will see with examples such as the Cantor set. We state an additional corollary:

Corollary 0.1. *If A is somewhere dense for some open $Y \subseteq X$, then $\overline{A \cap Y} = \bar{A} \cap \bar{Y}$*

We already have the inclusion that $\overline{A \cap Y} \subseteq \bar{A} \cap \bar{Y}$ (which is true for all sets). We prove the reverse inclusion. We see that:

$$\bar{Y} = \overline{A \cap Y} \wedge A \cap Y \subseteq A \implies \bar{Y} \subseteq \bar{A}$$

Thus $\bar{A} \cap \bar{Y} = \bar{Y} = \overline{A \cap Y} \supseteq \bar{A} \cap \bar{Y}$.

The Cantor Set

Properties

The Cantor Set is a nowhere dense, perfect set constructed by repeatedly removed the middle 1/3 interval of each interval. We denote each step of this iterative process as C_n where for all $I \subset C_{n-1}$, we remove their middle 1/3 portion, creating C_n.

An interesting property of the Cantor Set is that it is uncountable. For this, we examine a representation of the elements of the Cantor Set, called the ternary expansion:

$$\forall x \in C : x \coloneqq \sum_{i\in\mathbb{N}} \frac{2\sigma_i}{3^{i+1}}$$

where σ_x is a sequence in $\{0,1\}^{\mathbb{N}}$. This is called the ternary representation because it is in base 3. If a representation of a number in $[0,1]$ MUST contain a 1 in it, then that tells us it is in a middle 1/3 interval.

Based on its place in the decimal, such a ternary expansion tells us that in order to reach the number, we must pass through the interval $[1/3^n, (2/3)^n)$ (and endpoints remain in the Cantor Set), so it belongs in the middle interval.

This ternary expansion gives us an injection $f : \{0,1\}^{\mathbb{N}} \to C$. If $f(\sigma_1) = f(\sigma_2)$, then:

$$f(\sigma_1) = f(\sigma_2) \implies \sum_{i\in\mathbb{N}} \frac{2\sigma_{1i}}{3^{i+1}} = \sum_{i\in\mathbb{N}} \frac{2\sigma_{2i}}{3^{i+1}}$$

$$\implies \sum_{i\in\mathbb{N}} \frac{2(\sigma_{1i} - \sigma_{2i})}{3^{i+1}} = 0$$

Since 0 has a unique ternary expansion which is $0.00000..._3$ itself, then this means $\forall i \in \mathbb{N} : \sigma_{1i} = \sigma_{2i}$, proving equality of the sequences and injectivity of f.

Now because $\{0,1\}^{\mathbb{N}} \simeq \mathbb{R}$, then $\mathbb{R} \lesssim C$, so the Cantor set is uncountable (which will be an interesting result later on).

We note that each time the iterative process doubles the amount of intervals, so each C_n can be covered by 2^n closed balls of radius $3^{-n}/2$. This fact will be quite important when we examine integration over the indicator function on C. We first prove that the Cantor set is nowhere dense.

Lemma 0.3. *The Cantor Set is nowhere dense.*

First since each C_n is a closed set, then the construction of C_{n+1} involves removing only OPEN middle 1/3 sets, so C_{n+1} is also a finite union of 2^n closed sets, thus closed. C is also closed because it is the arbitrary union of closed sets C_n, so $\bar{C} = C$.

A point is in the interior if of a set A:

$$\exists \epsilon > 0 : B(x,\epsilon) \subseteq A$$

So we aim to prove the negation of this for all such points in the Cantor set. For all x in the Cantor set, since $x \in \cap_{n\in\mathbb{N}} C_n$, then $\forall n \in \mathbb{N} : x \in C_n$. We can find the following:

$$\forall \epsilon > 0, \exists n \in \mathbb{N} : \frac{1}{3^n} < 2\epsilon$$

Then we have that x belongs to some interval $I_{n'}$ in C_n. Because each interval is covered by a closed ball of radius $1/(2\cdot 3^n)$ centered at the midpoint (which may be x or not), then the ball $B(x,\epsilon)$ certainly exceeds the interval.

(If x is the midpoint, this follows from the covering. If x is not the midpoint, then the ball $B(x, 1/(2\cdot 3^n))$ already falls outside the interval, so we can extend it to the ϵ-ball by subset properties). Thus:

$$\forall \epsilon > 0, \exists y \notin C : y \in B(x,\epsilon)$$

Thus the Cantor set has no interior points and is nowhere dense.

Lemma 0.4. *The Cantor Set is perfect, which means it is closed and has no isolated points.*

Given some $x \in C$ and some $\epsilon > 0$, the previous lemma gave us the existence of $n \in \mathbb{N}$, for which x belongs to an interval in C_n. Once again by case work, we can show that at least one endpoint of the interval different from x should fall in the ball $B(x, \epsilon)$ and since endpoints remain in C, we have shown that no point is isolated:

$$\forall x \in C, \forall \epsilon > 0 : (B(x, \epsilon) \setminus \{x\}) \cap C \neq \varnothing$$

Integrability

We now examine the following function:

$$1_C := \begin{cases} 1 \ : \ x \in C \\ 0 \ : \ x \notin C \end{cases}$$

We aim to show this function is Riemann integrable. We first show that it is discontinuous at all points within the Cantor set (via limsup liminf criterion):

$$\forall x \in C, \forall \delta > 0 : \sup_{z:|z-x|<\delta} f(x) = 1 \wedge \inf_{z:|z-x|<\delta} f(x) = 0$$

The infimum part is true because of Lemma 0.3 (nowhere dense), which tells us that for any δ-ball around x, we can always find some point outside the Cantor set within the ball. The supremum can also be supported by the perfect + nowhere dense properties of C (if we consider the punctured open ball, perhaps to disprove existence of a limit). In that case, we have a discontinuity of second kind at each $x \in C$ as we can show both left and right limits do not exist.

Then since the above condition is true for all δ:

$$\lim_{\delta \longrightarrow 0+} \sup_{z:z \in B(x,\delta)} f(x) = 1 > \lim_{\delta \longrightarrow 0^+} \inf_{z:|z-x|<\delta} f(x) = 0$$

which tells us that f is not continuous at all $x \in C$. So now we have a function that is nowhere dense, with an uncountable number of discontinuities, that too of the second kind YET is Riemann integrable. Then:

1. Cardinality of the set of discontinuities while sufficient is too strong of a condition for Riemann integrability
   (a) Thomae's function in $[0, 1]$ showed that we can have countably many discontinuities (but of the first kind as limit exists at all rationals)
2. Type of discontinuities faces the same issue as well (1_C has discontinuities of the second kind)

Lemma 0.5. *The function 1_C is bounded and Riemann integrable on $[0, 1]$ and $\int_0^1 1_C dx = 0$.*

We note that by the Darboux definition, the lower integral over any partition is always 0. So we aim to show that the upper integral also converges to 0. We define the following:

$$\Pi_n := \{I_n \cup M_n : I_n \subseteq C_n\}$$

where I_n are the existing intervals after removing the middle 1/3 intervals denoted as M_n. Then:

$$\forall \Pi_n : U(f, \Pi_n) = \sum_{I_{n_i} \in I_n} 1 \cdot \Delta I_{n_i} + \sum_{M_{n_j} \in M_n} 0 \cdot \Delta M_{n_j}$$

This is because the supremum on I_n is always 1 (the endpoints are in C) and the supremum on M_n is 0 (it contains no points in the Cantor set). Then:

$$U(f, \Pi_n) = 1 \cdot \left(\frac{2}{3}\right)^n \implies \inf U(f, \Pi_n) = \lim_{n \to 0} U(f, \Pi_n) = 0$$

Thus the upper integral and lower integral agree at the value 0. The $(2/3)^n$ come from the fact that there are 2^n closed balls of radius $1/(2 \cdot 3^n)$ that cover all of C_n. Notice that the "length" of the Cantor set (the set of discontinuities in this case) goes to 0.